Practical 01: Standardisation of NaOH Using KHP

Electronic Laboratory Notebook (ELN) | ChemCollective Virtual Laboratory | Verified dataset release

1. Objective

To determine the apparent concentration of sodium hydroxide solution by titration against a known potassium hydrogen phthalate (KHP) solution using phenolphthalein indicator.

2. Experimental setup

- KHP concentration: 0.500 mol/L
- KHP aliquot: 10.000 mL
- Phenolphthalein indicator: 0.300 mL
- Initial flask volume: 10.300 mL
- Initial NaOH burette charge for every trial: 10.000 mL
- Reaction: $KHC_8H_4O_4 + NaOH \rightarrow KNaC_8H_4O_4 + H_2O$ (1:1)

3. Verified observations and results

Parameter	Trial 1	Trial 2	Trial 3	Unit	Basis
Endpoint colour	Pink	Pink	Pink	—	User-confirmed
Final flask volume	15.800	15.600	15.600	mL	User-confirmed
Initial burette charge	10.000	10.000	10.000	mL	Every trial
Final burette reading	4.500	4.700	4.700	mL	User-confirmed
NaOH titre	5.500	5.300	5.300	mL	Initial – final burette reading
Endpoint pH	12.09	10.57	10.57	pH	Source summary

4. Calculations

For the 1:1 reaction: M(KHP) × V(KHP) = M(NaOH) × V(NaOH). Therefore M(NaOH) = (0.500 × 10.000) / titre.

Trial 1: titre = 10.000 − 4.500 = 5.500 mL; M(NaOH) = 5.000 / 5.500 = 0.9091 mol/L.

Trial 2: titre = 10.000 − 4.700 = 5.300 mL; M(NaOH) = 5.000 / 5.300 = 0.9434 mol/L.

Trial 3: titre = 10.000 − 4.700 = 5.300 mL; M(NaOH) = 5.000 / 5.300 = 0.9434 mol/L.

Mean apparent concentration = 0.9320 mol/L; sample standard deviation = 0.0198 mol/L; %RSD = 2.13%.

5. Endpoint and data-integrity notes

All three endpoint observations were pink. Trial 2 endpoint was at a final flask volume of 15.600 mL. The logger label “54” refers to its reading number; it must not be interpreted as 54 actual additions of 0.100 mL. The volume balance and burette readings establish a titre of 5.300 mL.

Trial 3 endpoint pH was confirmed by the user as 10.57. Endpoint pH values: Trial 1 = 12.09; Trial 2 = 10.57; Trial 3 = 10.57. The relatively high endpoint pH values are disclosed as a possible endpoint overshoot/simulator-response limitation.

6. Conclusion

The three virtual titration trials give apparent NaOH concentrations of 0.9091 M, 0.9434 M and 0.9434 M, with a mean of 0.9320 M and %RSD of 2.13%. These are calculated results from the recorded virtual endpoint titres. The dataset is suitable for a documented virtual analytical chemistry case study, but it is not a certified standardisation or evidence of physical wet-laboratory execution.

7. Method and documentation limitations

The experiment was performed in ChemCollective Virtual Lab. No independent calibration certificate, physical instrument qualification, or physical laboratory measurements are available. The term “apparent concentration” is therefore used throughout.